

Power Spectral Density and the Periodogram

From second-order statistics to finite-data spectral estimates

Learning objectives

ROADMAP

0:00–0:03

Connect autocorrelation and PSD through the Wiener–Khinchin theorem.

Predict how an LTI filter reshapes the PSD of a WSS input.

Compute and interpret a finite-record periodogram.

Explain why a raw periodogram is noisy even when N is large.

Separate frequency-grid refinement from statistical variance reduction.

Core question

What can one finite realization tell us about the underlying distribution of signal power over frequency?

PSD: “where does the average power live?”

WIENER-KHINCHIN

0:03–0:09

For a WSS process, the autocorrelation depends only on lag m :

$$r_{xx}[m] = E\{x[n] x^*[n+m]\}$$

The power spectral density is its DTFT:

$$S_{xx}(e^{j\omega}) = \sum_m r_{xx}[m] e^{-j\omega m}$$

Interpretation

Correlation describes similarity across time lag. The PSD describes the same second-order information organized by frequency.

Wiener–Khinchin in both directions

WIENER-KHINCHIN

0:09–0:14

Forward transform

$$S_{xx}(e^{j\omega}) = \text{DTFT}\{r_{xx}[m]\}$$

Inverse transform

$$r_{xx}[m] = (1/2\pi) \int_{-\pi}^{\pi} S_{xx}(e^{j\omega}) e^{j\omega m} d\omega$$

At $m = 0$

$r_{xx}[0] = E\{|x[n]|^2\}$ = average power =
area under the PSD / 2π .

Units matter

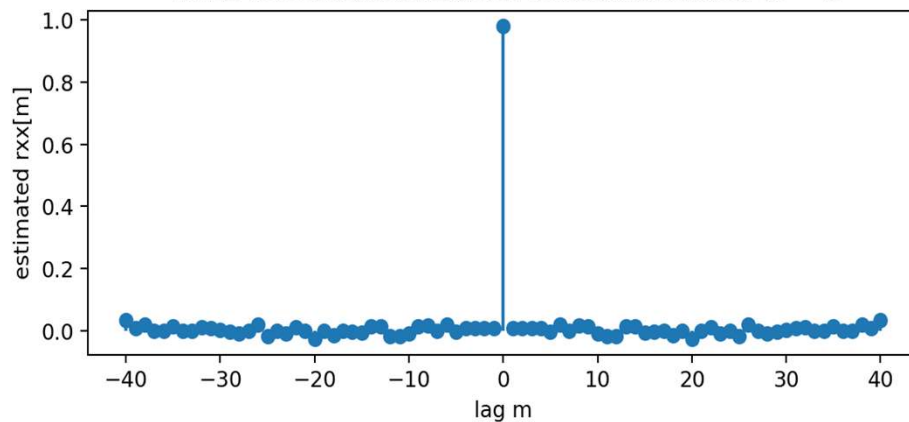
PSD is power per unit frequency.
Changing from rad/sample to Hz
changes the numerical scaling.

Example: white noise

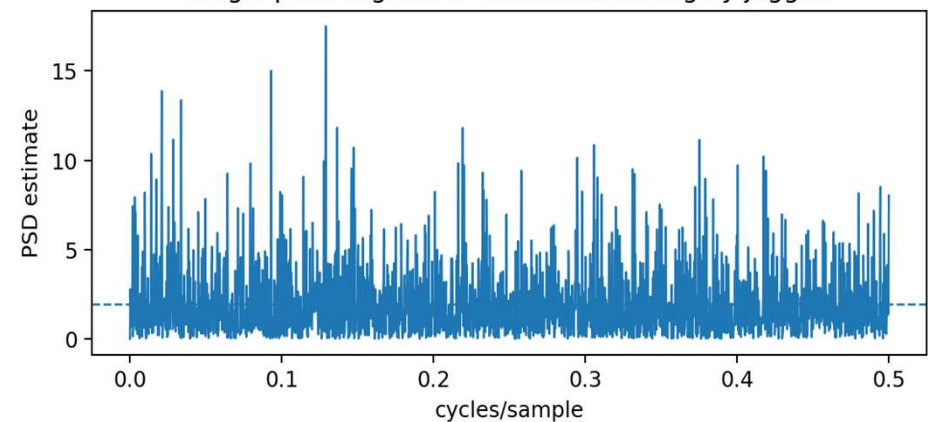
WIENER-KHINCHIN

0:14–0:20

White noise: autocorrelation concentrated near $m = 0$



Single periodogram of white noise is highly jagged



$$\text{Theory: } r_{xx}[m] = \sigma^2 \delta[m] \Leftrightarrow S_{xx}(e^{j\omega}) = \sigma^2$$

Important

A flat theoretical PSD does not mean a single periodogram will look flat. The estimator itself has large random fluctuations.

PSD through an LTI system

FILTERING

0:20–0:27

$$y[n] = h[n] * x[n]$$

For a WSS input and stable LTI system:

$$S_{YY}(e^{j\omega}) = S_{XX}(e^{j\omega}) |H(e^{j\omega})|^2$$

The filter multiplies amplitude spectra by H .

Power is proportional to squared magnitude, hence $|H|^2$.

Phase affects waveform relationships, but the output PSD depends on $|H|^2$.

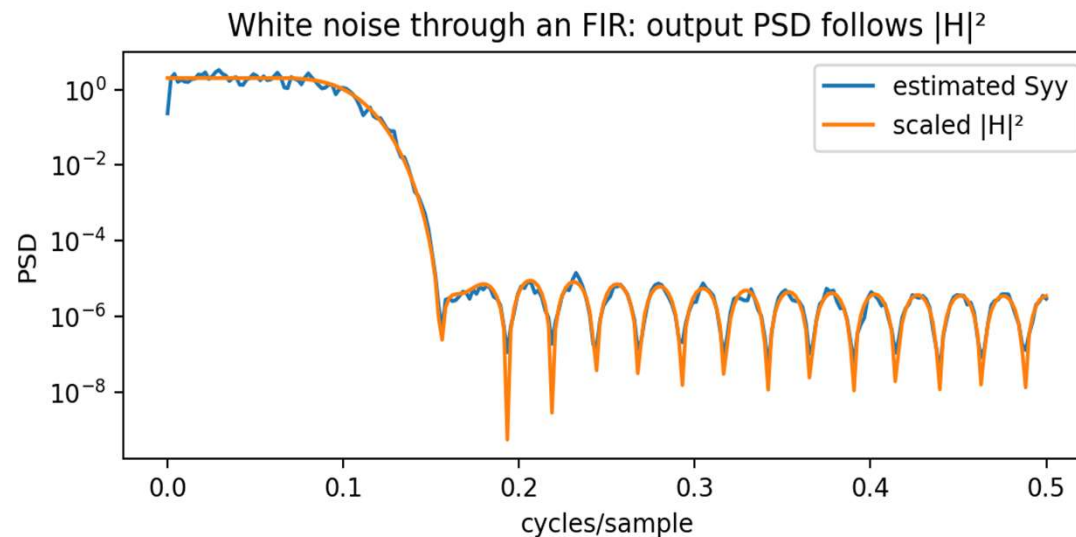
White-noise special case

If $S_{XX} = \sigma^2$, then $S_{YY} = \sigma^2 |H|^2$.
Filtering “colors” white noise.

Numerical check: filtering colors white noise

FILTERING

0:27–0:35



What to notice

The random estimate fluctuates around the deterministic shape $|H|^2$. The filter imposes spectral structure on otherwise white input noise.

Concept check

If H has a deep notch at ω_0 , what should happen to S_{yy} at ω_0 ?

Good answer: the output PSD should also have a deep notch, scaled by the input PSD.

From theory to finite data

PERIODOGRAM

0:35–0:39

Theory assumes ensemble statistics such as $E\{\cdot\}$.

In practice we often have one finite record:

$x[0], x[1], \dots, x[N-1]$

Question: how do we estimate S_{xx} from those N samples?

Key change

PSD is a property of the stochastic process. A spectral estimate is a random quantity computed from finite data.

The periodogram

PERIDOGRAM

0:39–0:45

Compute a finite-length Fourier transform and square its magnitude:

$$I(\omega) = (1/N) \left| \sum_{n=0}^{N-1} x[n] e^{-j\omega n} \right|^2$$

FFT evaluates this expression efficiently on a frequency grid.

Normalization conventions differ: always check one-sided/two-sided and density/spectrum scaling.

For real data, plotting only $0 \dots F_s/2$ is common.

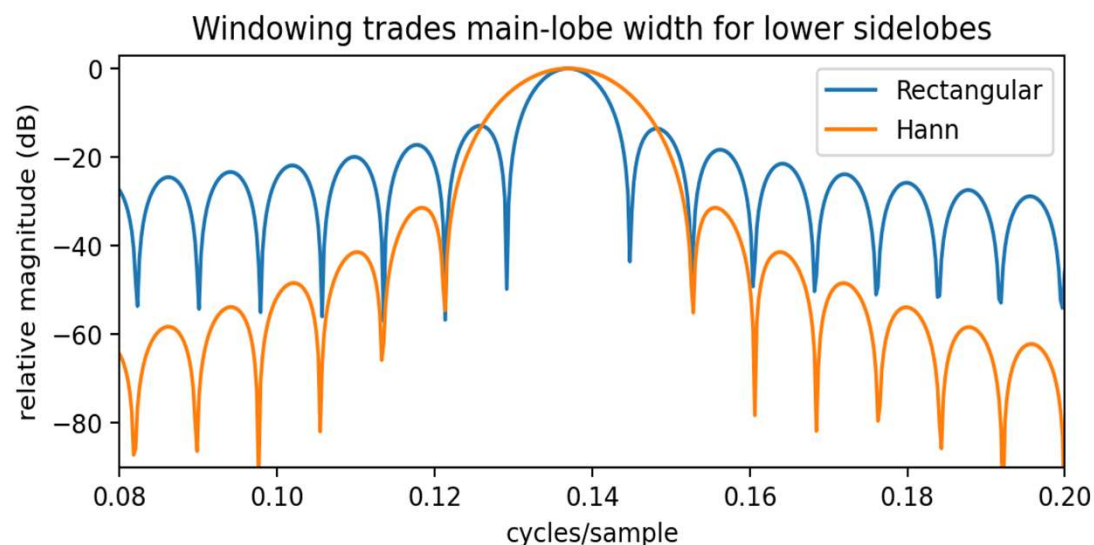
Mental model

Periodogram = squared magnitude of a finite-record spectrum, normalized to estimate power.

Why a tone leaks into nearby bins

PERIODOGRAM

0:45–0:50



Finite record = windowing

Observing only N samples multiplies the infinite process by a finite window. In frequency, that becomes convolution with the window spectrum.

Trade-off

Tapered windows reduce sidelobes (leakage) but widen the main lobe, reducing the ability to separate nearby tones.

Resolution: what increasing N really improves

PERIODOGRAM

0:50–0:55

FFT-bin spacing for N samples at F_s :

$$\Delta f = F_s / N$$

Longer records give a finer frequency grid.

Longer observation also narrows the rectangular-window main lobe.

But a raw periodogram does not become statistically smooth just because N is large.

Common misconception

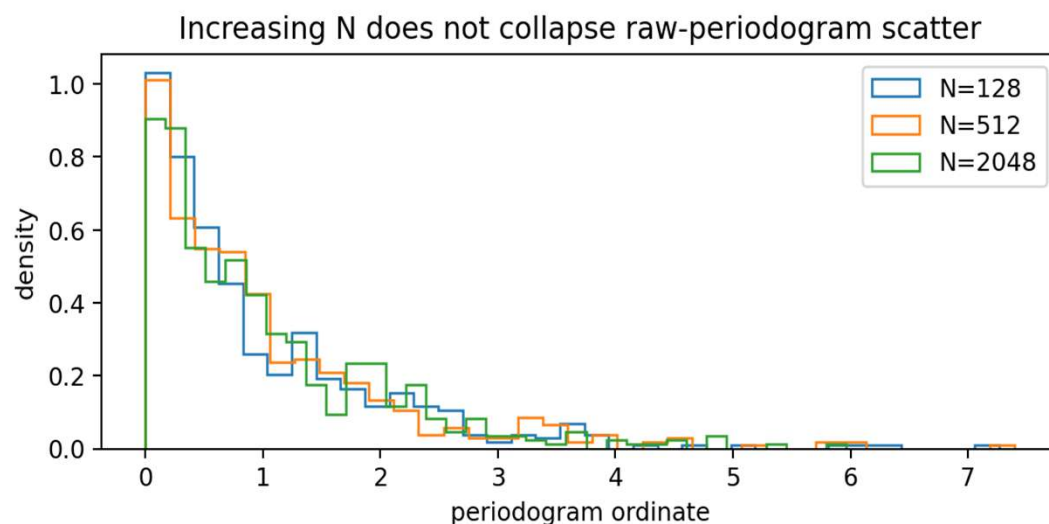
“More FFT points” can mean zero-padding, which only interpolates the plotted spectrum. It does not create new information.

Why the raw periodogram is inconsistent

STATISTICAL FLAW

0:55–1:03

Consistency requires the estimator variance to vanish as $N \rightarrow \infty$.



White Gaussian noise intuition

At a non-DC Fourier bin, the DFT coefficient remains random. Its squared magnitude is approximately exponential, so relative scatter stays large.

For an idealized white-noise bin:
 $E\{I\} \approx \sigma^2$, $\text{Var}\{I\} \approx \sigma^4$

Coefficient of variation ≈ 1 — it does not shrink with N .

What “inconsistent” does—and does not—mean

STATISTICAL FLAW

1:03–1:08

It does mean

The raw periodogram at a fixed frequency remains noisy across repeated realizations; its variance does not generally tend to zero.

It does not mean

The periodogram is useless. It can reveal strong tones, and it is the building block for lower-variance estimators such as Bartlett and Welch.

Estimator design question:

Can we trade some frequency resolution for lower variance by averaging?

That is exactly what Lecture 18 does.

Concept check: two records of the same process

CONCEPT CHECK

1:08–1:12

Two long records come from the same zero-mean WSS process. Their raw periodograms look noticeably different.

Is that evidence that the process PSD changed?

Good answer

Not necessarily. A periodogram is itself random and has high variance. We need averaging or confidence reasoning before attributing differences to the underlying process.

Coding demo: four things to verify

DEMO

1:12–1:14

1. Estimate white-noise autocorrelation and compare its PSD with a flat theoretical spectrum.
2. Filter white noise and verify S_{yy} follows $|H|^2$.
3. Repeat many periodograms and observe that the scatter remains large as N grows.
4. Change only the FFT zero-padding and distinguish smoother plotting from true resolution.

Notebook

Week9_Spectral_Estimation_DSP_ML_Demos.ipynb reproduces these experiments with NumPy/SciPy.

Lecture 17 wrap-up

WRAP-UP

1:14–1:15

Wiener–Khinchin: PSD and autocorrelation are a DTFT pair.

LTI systems reshape PSD by $|H|^2$.

A periodogram is a finite-record PSD estimate based on $|FFT|^2$.

Longer records improve frequency detail, but raw-periodogram variance remains high.

Averaging periodograms is the next step.

Exit ticket

Why does a longer record not automatically make a raw periodogram a low-variance PSD estimator?